Introduction of Sustainability and Environmental Ethics

WI1103 Introduction of Environmental Principles (Week - 1)



Course materials

Week	Material
1	The concept of human life in the Earth's environment
2	The impact of human behavior on the environment
3	The Earth and overpopulation vs. overconsumption
4	Similarities: poverty, climate change
5	Introduction to the SDG concept
6	Continuation of the SDGs and the global challenge of climate change; net zero carbon
7	Types of solutions are nexus-based, including resource efficiency, environmental impact, and resource reuse
9	F/S material: sustainability interpreted in their respective fields
10	Project introduction, literature review related to the group assignment topic
11	Preparation for data collection for observations, interviews, and/or questionnaires
12	Data collection: observations, interviews, and questionnaires
13	Analysis of field data collection results
14	Discussion of proposed solutions based on student knowledge
15	Finalization of project activities

Report/booklet/logbook/PPT Grading

Weight	Aspect	4
10 %	Communication	These four aspects are met: The document looks neat The explanations in the document are easy to understand There are no errors in the writing The explanations in the document are complete
10 %	Ethics	These four aspects are met: Ethical considerations are evident in the analysis The language used is polite The use of appropriate citations and reference lists There is no indication of plagiarism in the assignment
20 %	Cooperation	These four aspects are met: The document is submitted on time The document is formatted appropriately There are no questions regarding the implementation of the document assignment outside of class The document is submitted in the appropriate place
40 %	Problem identification	These four aspects are met: The problem raised is a real problem from the literature review and data collection The problem identified is indeed a problem related to sustainability The analysis to conclude the problem is carried out appropriately The coherence of the analysis is carried out well
20 %	Solution proposal	These four aspects are met: The solution is able to address the identified problem, although not completely It is apparent that this solution is the result of problem identification and not a predetermined solution and then searching for the problem The solution offered takes sustainability into account The solution offered is not haphazard

Popular Media Grading: poster/comic/short video

Weight	Aspect	4
30%	Communication	These four aspects are met: The media looks engaging The explanations in the media are easy to understand The media quality is good The language used in the posts is good
30%	Coopration	These four aspects are met: Assignments are submitted on time Media creation requirements are met There are no questions regarding the implementation of posting assignments outside of class Media is collected in the appropriate place
30%	Learning	These four aspects are met: The information presented in the media is coherent There is a match between the problem raised and the solution offered The information presented in the media is concise and to the point Sustainability analysis is visible in the posts
10%	Ethics	These four aspects are met: Ethical considerations are visible in the analysis The language used is polite The information does not trigger conflict There is no indication of plagiarism in the assignment

Learning Outcomes

CPMK 1 Learning Skills

1. Students are able to understand the concept of sustainability from a holistic and/or philosophical perspective that integrates environmental/earth ethics considerations, interdisciplinary approaches, and personal reflection.
2. Students are able to understand the uncertainty and ambiguity in identifying sustainability issues, yet develop anticipatory thinking in implementing them.
3. Students are able to explain the relationship between the concepts of the built environment/infrastructure/technology and natural systems/embodied environment as a link or nexus.
4. Students are able to explain the challenges of development in the global south in applying the concept of sustainability, especially through the sustainable development goals.
5. Students are able to identify sustainability issues and offer alternative perspectives that are relevant in the local context.

CPMK 2 Collaborative Skills

1. Students are able to work collaboratively in teams, particularly across disciplines, to describe the context in which sustainability issues are understood.

CPMK 3 Communication skills:

1. Students are able to communicate their ideas in the form of documents and popular media.

CPMK 4 Ethical application skills:

1. Students are able to apply ethics related to sustainability and learning ethics when involved in decision-making.

Evaluation

Grading Component	Description	Weighting (%)
Individual online quizzes	Online quizzes with questions from the central team. Work done Friday, 5:00 PM - Monday, 8:00 AM.	10
Group class assignments	Group class assignments. Handed in to each faculty/student. Conducted at least once.	10
Mid-term exams	Written mid-term exams in class.	30
Activity reports	Written documents detailing student work. This can be in the form of logbooks, detailed PPTs, reports, or other relevant formats.	35
Popular media	Explanations of major assignments in popular formats. This can be posters, short videos, social media posts, comics, or other relevant media.	15

Course Activities

- Classroom lectures
- Asynchronous materials in the form of slides and reference videos
- Online quizzes
- Midterm and final exams are held centrally

Course Activities

- Group assignments in class
- Project assignments
- Project outputs include reports and popular media
- Groups consist of 5-6 students

Week 1

1. What is sustainability? Definitions and concepts of sustainability
    1. What is the definition of sustainability?
    2. When did sustainability issues begin to be discussed?
    3. What are the dimensions of sustainability?
    4. Environment and ecosystems
    5. Human impacts on ecosystems
    6. Ecological law
2. Environmental ethics
    1. Illustration of environmental ethics: animal agriculture
    2. Dominion thesis and related theories
    3. Conservation dilemma scenarios

What is Sustainability?

Definitions and Concepts

What is *sustainability?*

Can sustainability be scientifically defined?

Sustainability is open-ended and temporary

Sustainability should be understood as an open-ended, positive and temporary concept. It is open-ended because the ***social understanding*** of sustainable development depends on time and situation, as well as culture and knowledge.

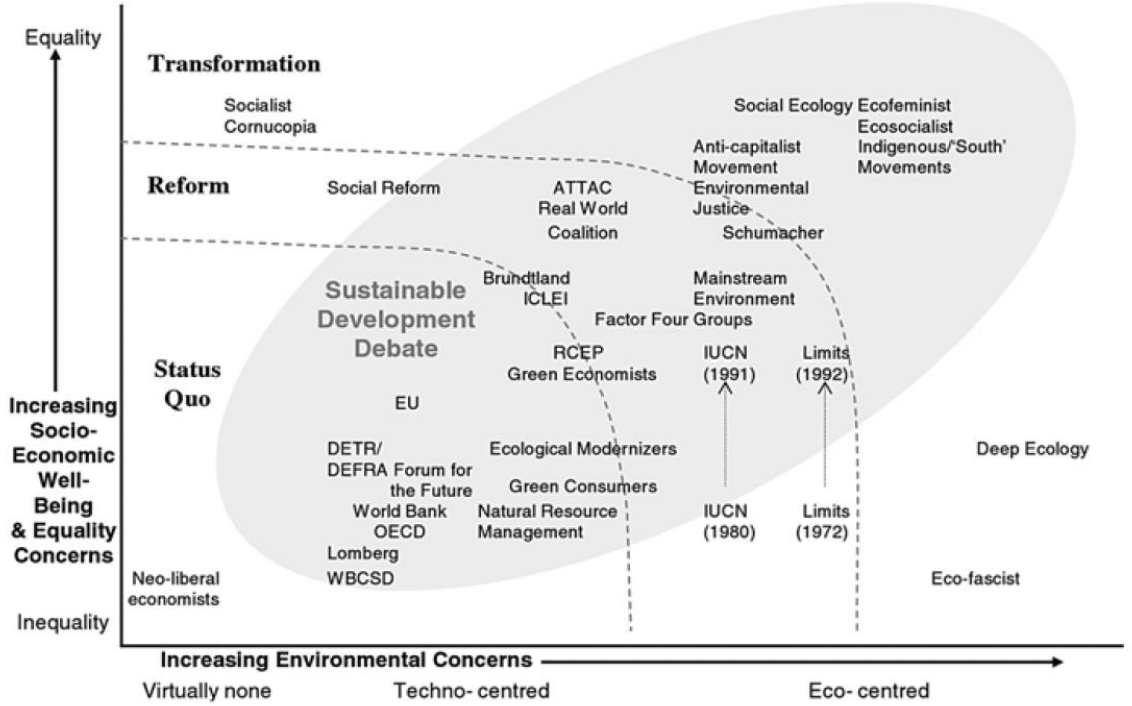


Fig. 2.3 Mapping different views of sustainable development (Hopwood et al. 2005: 41). The linking of justice criteria and environmental concerns illustrates the wide range of approaches regarding their normative priorities and so provides a useful basis for the critical evaluation of different interpretations and constructs in sustainability discourse

When did the concept of sustainability start being discussed?

1713

Carl von Carlowitz, German mining director
Calls for “continuous, stable and sustainable utilization of forests”.

1850

Thomas Maltus, economist
He identified imbalances between the resources in a habitat and its population. He predicted famines, epidemics, and wars. His work is cited as the first systematic study of the ecological limits to growth in the world and an early source of sustainability.

1960

Boulding, Ayres & Kneese, Georgescu-Roegen, Daly, economists
Due to the many environmental disasters that have occurred, environmental issues have become a public concern. Economists are putting the concept of nature and environment back on the economic agenda.

1983

The Brundtland Comission, World Commission on Environment and Development (WCED)
Published Our Common Future which details the definition of the concept of sustainability development. The report explains that environment, economy, and society are interdependent and interconnected concepts.

1994

The German Advisory Council
Divides the three elements of sustainability development:
1. Humanity's responsibility for its natural environment
2. Humanity's responsibility to the social world
3. Humanity's responsibility to itself

2003

Sustainability concepts can be divided into unidimensional and **multidimensional.**

What are the dimensions of sustainability?

- **Economic**
  The ability of an economy to meet the needs of the present without compromising the ability of future generations to meet their own needs.

- **Environment**
  the ability of the environment to support human life and well-being in the long term

- **Social**
  the ability of society to meet the needs of its members and ensure that everyone has the opportunity to lead a healthy, productive and fulfilling life.

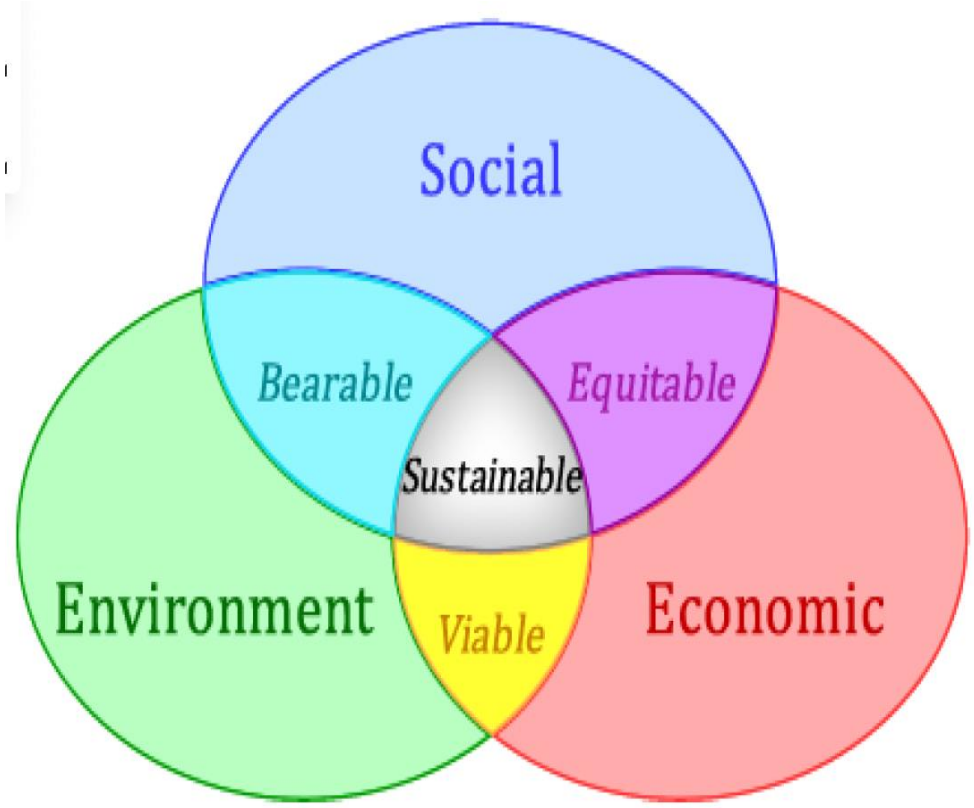
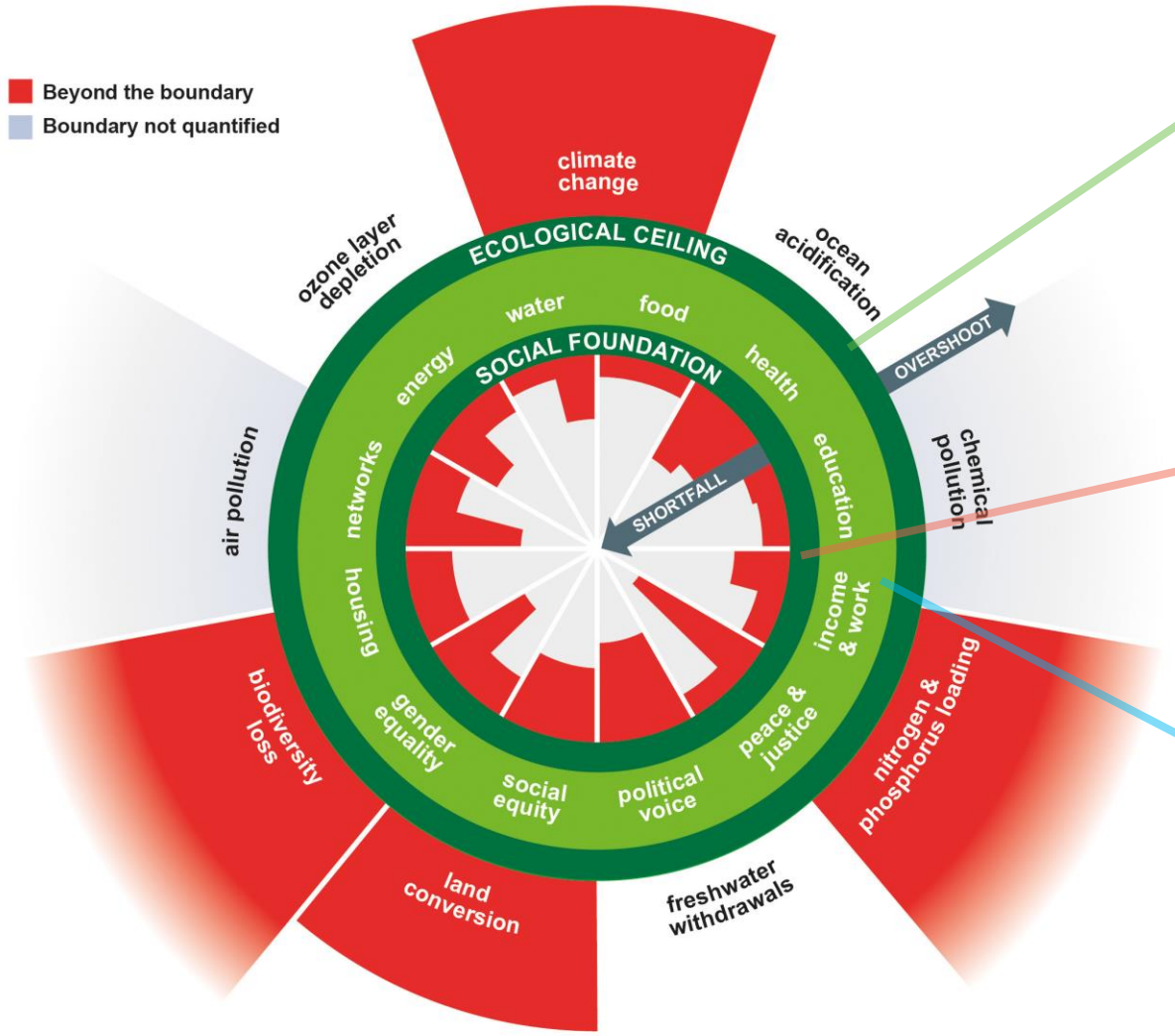


Figure 1.1: Overlapping Themes of the Sustainability Paradigm A depiction of the sustainability paradigm in terms of its three main components, showing various intersections among them. *Source: International Union for the Conservation of Nature*[7]

Doughnut Economics



The Outer Circle – Ecological Ceiling

- ← Living within our means
- ← Minimizing the harm we do to the natural environment
- ← Maximizing the amount of ecological benefit we give back

The Inner Circle – Social Foundations

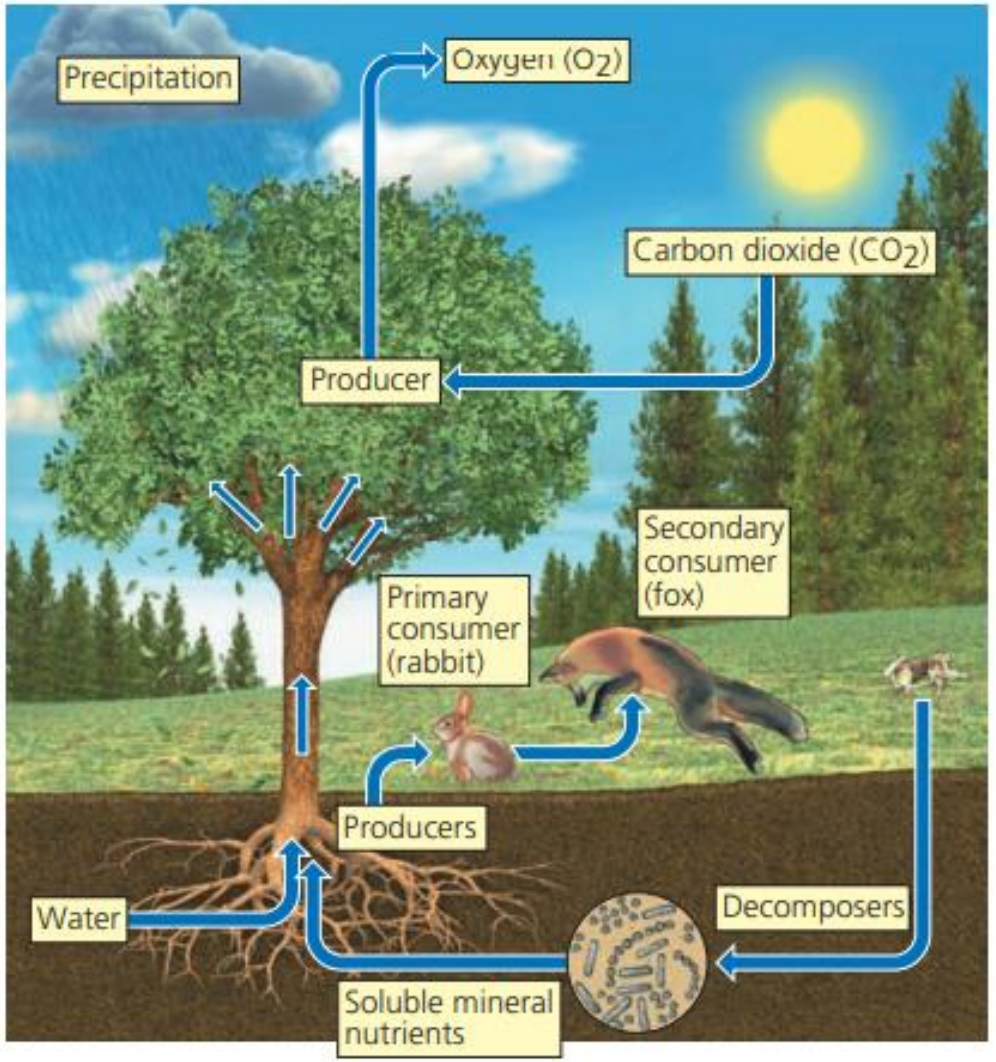
- ← Addressing shortfalls and meeting needs
- ← Creating opportunity for all

The Centre – Our Aim

- ← The "sweet spot" where we are meeting the needs of all, while respecting the limits of our environment
- ← Where we will thrive

Environment

- The environment is foundational; societies and economies depend on it and must operate within planetary boundaries.
- The environment is everything around us. It includes all living things (such as plants and animals) and non-living things (such as air, water, and sunlight) that interact with us.
- One key component of environmental science is ecology, which examines ecosystems.
- An ecosystem is a functional unit consisting of a community of living organisms (biotic) and the environment (abiotic), interacting and connected through energy flow and nutrient cycling.



Komponen utama biotik (makhluk hidup) dan abiotik (tidak hidup) dari suatu ekosistem (Miller & Spoolman, 2019)

Ecosystem

- An ecosystem can encompass a large area, such as a lake, forest, or island, or even the space beneath a fallen log or a spring in the desert.
- Many ecologists view the entire biosphere as a global ecosystem, the sum of all local ecosystems on Earth.
- This means:
  - The entire surface of the Earth is a network of interconnected ecosystems.
  - All activities within an ecosystem will affect various processes within the ecosystem (material cycles and energy flows).



Ekosistem akuatik, terumbu karang, *Papua New Guinea, Photograph by Karen Schofield* (https://education.nationalgeographic.org/resource/ecosystem/)



Ekosistem terestrial, Gurun Namibia, *Guinea, Photograph by Dagmara Sperling* (https://education.nationalgeographic.org/resource/ecosystem/)

Social: The effect of human activities to ecosystem/ environment

- Ultimately, all activities will impact the ecosystem/environment in which we live.
- Examples:
- The clearing of forest land for agriculture.
- The fishing activity at sea.
- The cutting down of trees for timber.
- Etc.

- Human activities alter energy flows and nutrient/water cycles, changing the structure of ecosystems—ultimately shifting the ecosystem's functions that support society and the economy.

BIODIVERSITY, CONSERVATION | CRISIS - BIOSYSTEM VIABILITY

95% of the Earth's Soil on Course to Be Degraded by 2050

BY SAVE SOIL | GLOBAL COMMONS | JUN 17TH 2024 | 3 MINS

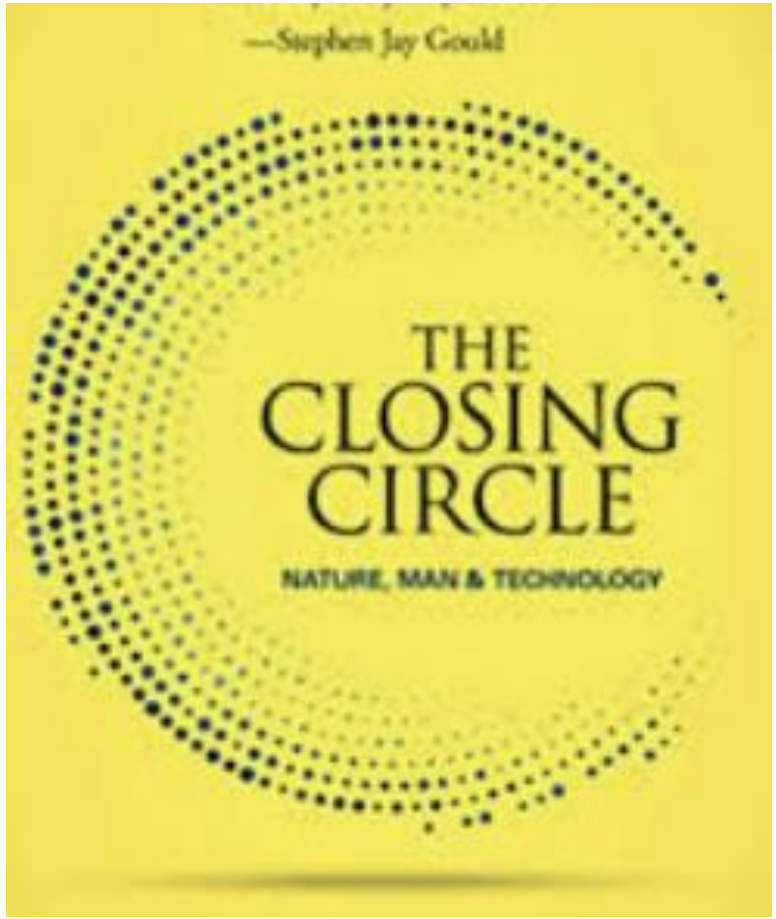
EARTH.ORG IS POWERED BY OVER 150 CONTRIBUTING WRITERS



Scientists warn that 24 billion tons of fertile soil is being lost per year, largely due to unsustainable agriculture practices.

Law of Ecology:

1. Every element is connected to everything else.
2. Every species depends on every other species.
3. Everything must go somewhere.
4. The laws of thermodynamics apply.
5. Nature Knows Best.
6. There is no such thing as a free lunch.



Environmental Ethics

Theories

Environmental Ethics

Theories

How do you view the efficiency of the livestock industry?

Farrowing crates are metal crates with bars that are placed inside the pig barn. These crates are made to house pregnant sows just before giving birth until 30-35 days postpartum. This method restricts the movement of pigs during and after labor.

This method is known to be more **economical** as it is thought to reduce the likelihood and accidental mortality of piglets thereby increasing production and economic returns.



Opponents of Agricultural Industrialization

Argument A

Animals are not “capital”, but sentient beings.

Argument B

Shutting down small and traditional livestock/farming businesses

Pollute water with the potential to disrupt human health downstream

Arguments A and B = ethics, the branch of philosophy that questions what is right and what is wrong; morality. But there is a difference. **Which one is human-centered/anthropocentric?**

Role of Ethics

Throughout history, codes of conduct have provided guidance to civilizations, including some philosophies and religions.

In the ethical tradition of “Western” civilization => **One's actions** and their impact on others



People of Color Breathe More Hazardous Air. The Sources Are Everywhere.

Researchers uncovered stark disparities between white people and minorities across thousands of categories of pollution, including trucks, industry, agriculture and even restaurants.

balairung press

REDAKSI NALAR

WAWASAN

Ketidakadilan Lingkungan di Tengah Pusaran Pembangunan Infrastruktur

Mei 10, 2022

Dominion Thesis

Humans are the pinnacle of creation: as such, humans are given the ethical freedom to use nature in any way they deem beneficial.

Humans are a separate, and superior, entity to nature and its contents.

Nature is only useful as long as it has benefits for humans



17th century wolf eradication program >> First environmental regulations passed in North America

Related Theories

John Locke

The fundamental of liberalism

Trust the government, but limited to protect individual liberty => property rights

Nature as a utility, a useful “property” for human



Anthropocentrism
Ethical viewpoint that humans are the primary factor in determining whether actions toward nature are right or wrong.

Utilitarianism
A normative ethical theory that determines the value of something is measured based on its benefits (maximizing individual/societal welfare).

Stewardship

Moral responsibility to protect the environment and living things

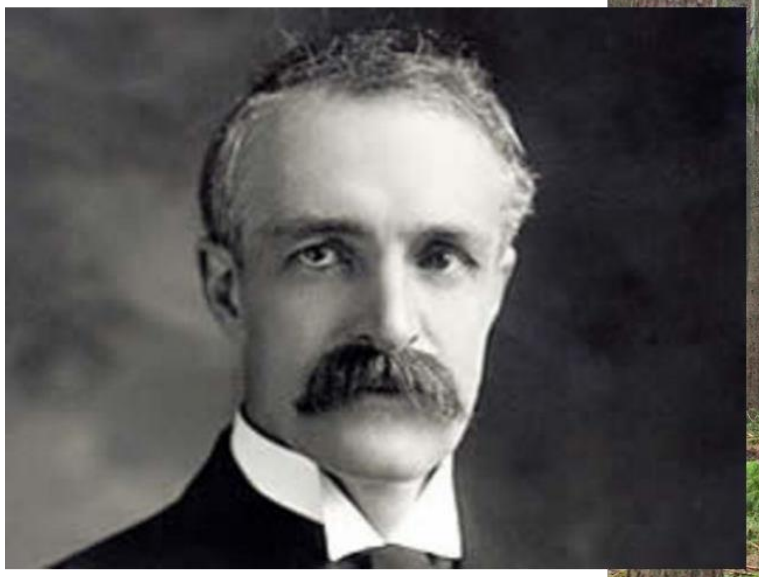


Related Theories

Cifford Pinchot

Conservationist => efficient and sustainable use of natural resources; *“the greatest good for the greatest number*”

Enemy of the ‘*timber baron*’

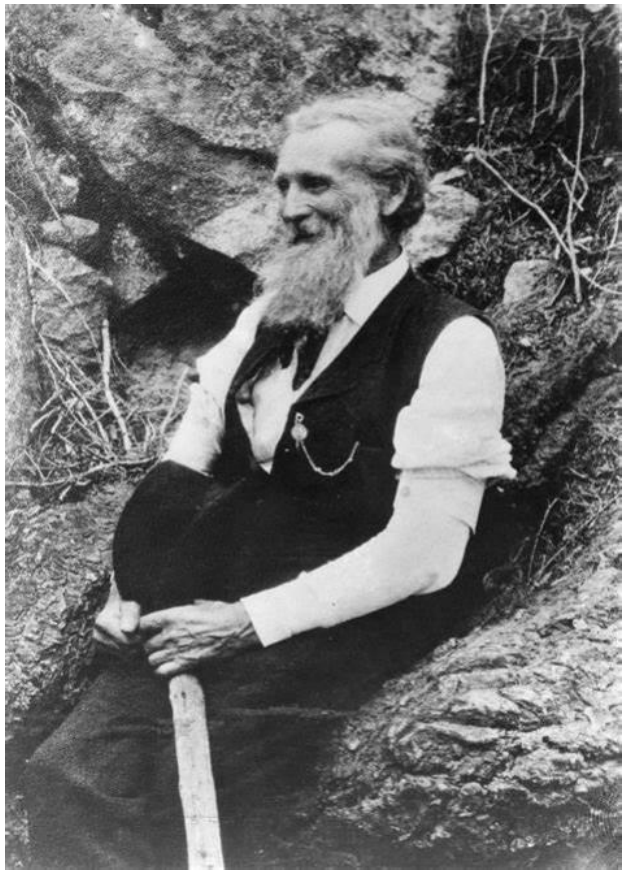


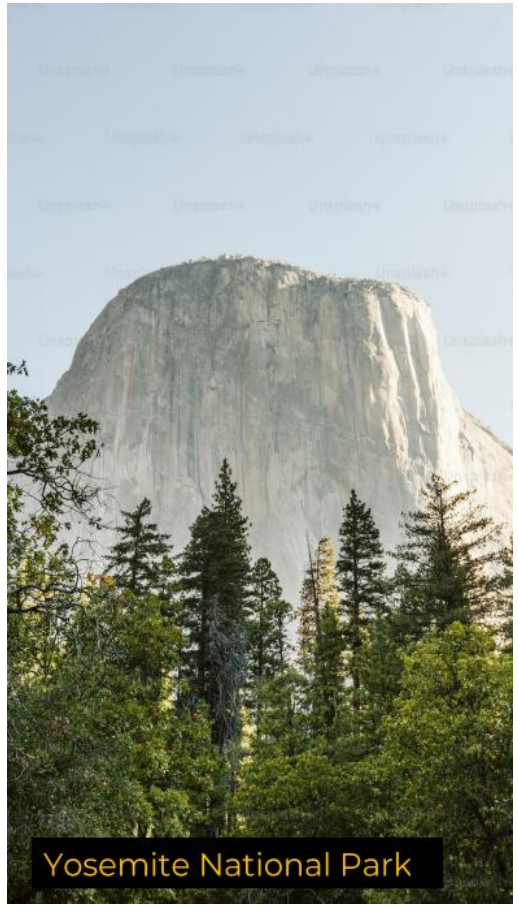


Related Theories

John Muir

- Preservationist
- Management of natural resources primarily for protection and preservation, for the benefit of wildlife
- Non-utilitarian
- Pioneer of conservation parks in the United States







Hetch Hetchy Valley

Pinchot vs muir

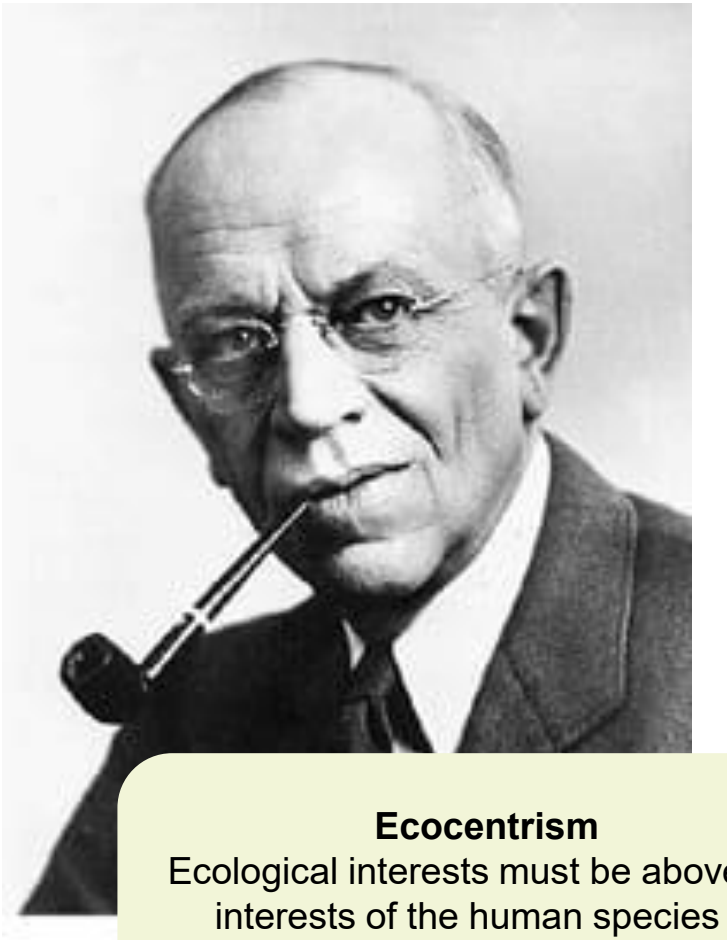
Dams provide greater benefits to society

HHVs are biodiversity 'treasures' and dams disrupt wildlife

Related Theories

Aldo Leopold "*Land Ethic*"

- Land is not just 'property'
- The concept of 'interdependence' in society: to survive and be part of society, individuals must give up their absolute freedom voluntarily.
- Leopold stretches the boundaries of the concept of dependency by including nature and other living things in it
- **Moral extension "we must change our relationship with nature... from conqueror... to plain member and citizen" (Leonopold, 1949)**



Ecocentrism
Ecological interests must be above the interests of the human species in determining whether an action is right or wrong.

Conservation Dilemma Scenario

A watershed is an important location for migratory salmon species in their life cycle. The watershed is also home to small communities of indigenous people who have depended on salmon for generations for their livelihoods.

An energy company proposed a dam construction project. The dam will bring significant economic benefits, including national electricity supply, employment, and more. However, the dam would directly threaten salmon migration routes, significantly reduce salmon populations and disrupt indigenous peoples' way of life.

The government supported the development as it was in line with regional and national economic growth plans, but there was strong opposition from environmentalists and some members of the indigenous community.